HEFTNet: modelling the di-Higgs m_{hh} spectrum and its uncertainty

P. Bhat (Fermilab), N. De Filippis (Politecnico di Bari),
K. F. Johnson, H. B. Prosper, R. Starnes (Florida State University)

Executive summary and roadmap, October 31, 2026

Goal. A fast surrogate for POWHEG: the di-Higgs cross section as a function of m_{hh} (255–1455 GeV) and of five HEFT coefficients, with a calibrated uncertainty, written as $d\sigma/dm_{hh} = \sum_{i=1}^{23} C_i(\theta)\, a_i(m_{hh})$ with known polynomials C_i and 23 functions a_i of m_{hh} alone. Data: 3,989 POWHEG spectra for the fit; 1,000 independent test spectra for validation.

Principal findings

1. **The primary model is the smooth** 23×80 **model**: each $a_i(m_{hh})$ is a smoothed spline times the phase-space factor, fitted to the bin *integrals* by Poisson maximum likelihood, with smoothness set by the evidence. It is linear in its 920 parameters, so its posterior is exactly Gaussian; no training, a minute on a laptop. It is a genuine density in m_{hh}: fitted on 30-GeV bins it predicts the 15-GeV split to within the noise above 360 GeV.
2. **It is as accurate as any model of this form can be.** A free fit of 23 numbers per bin, the floor for the form $\sum_i C_i a_i$, does no better (deviance per bin 1.56/1.10 below/above 555 GeV for both); neither does the network ensemble (1.56/1.12).
3. **The data are exact Poisson counts**, $q_s\times$ integers (40,000 events per run), and weighting by the *observed* counts biases fits low; weighting by the model's Poisson variance (iterated) removes the bias.
4. **The residual model error comes from cancellations.** Spectra whose cross section is a small difference of large terms deviate smoothly from the 23-term form, in POWHEG itself. A discrepancy $\varepsilon(m_{hh}) \sum_i |C_i a_i|$, with $\varepsilon = 0.08\%$ at threshold to 3.7% at 1.45 TeV, fitted on rows not used in the fit, gives pull s.d. 1.004 (1.011/1.001 below/above 555 GeV), 95.4% within ± 2, and calibration at every degree of cancellation. The fit uncertainty itself is 2.9% of the noise.
5. **Cross-checks agree.** A 20-network ensemble (L1 + Huber-pull fine-tune) gives the same accuracy and, with the same discrepancy, the same calibration; its spread is 80% optimisation. A Bayesian network by HMC undersampled (0.5% of the noise); a mixture of Laplace approximations gives 2.4% $\oplus$ 5.6%, consistent with the smooth model.

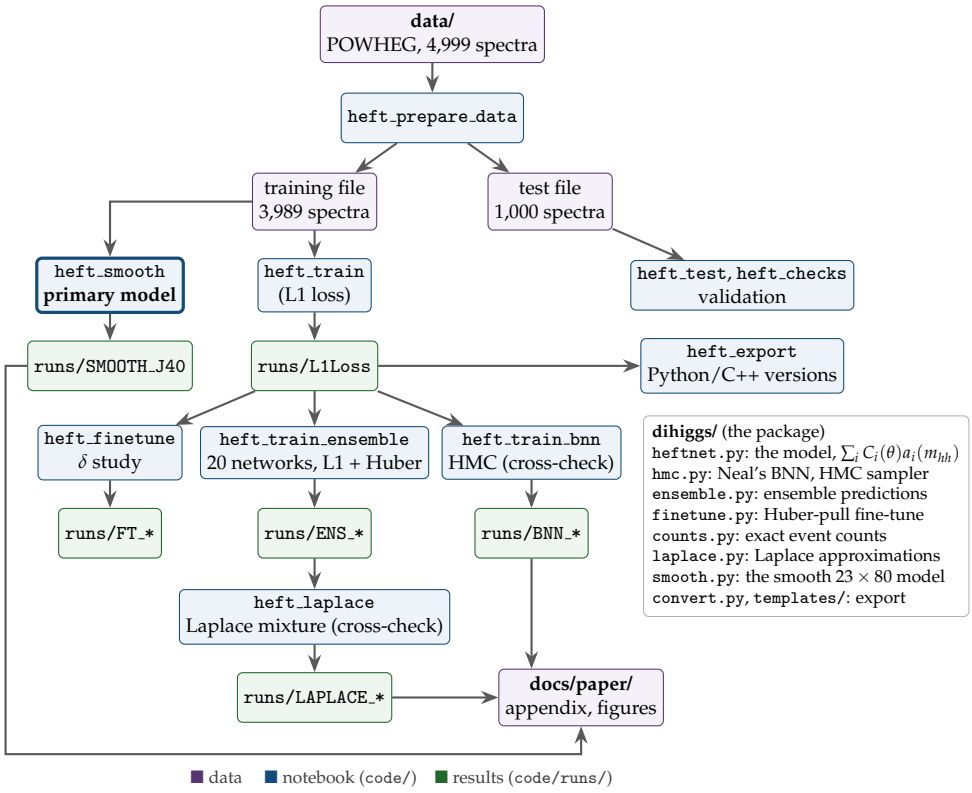
Investigations, in order.

- *BNN with HMC* (`hmc.py`); diagnosis in `docs/bnn/bnn_plan2.pdf`.
- *Ensembles* (`ensemble.py`): seeds vs bootstrap, reference ensembles (`docs/ensemble`).
- *Huber-pull fine-tune* (`finetune.py`), `docs/huber`.
- *Exact counts* (`counts.py`), `docs/counts`.
- *Mixture of Laplace approximations* (`laplace.py`), `docs/bnn/laplace_explained.pdf`.
- *Origin of the model error*: free-fit benchmark, cancellation-aware discrepancy (`docs/modelerror`).
- *Smooth* 23×80 *model* (`smooth.py`), `docs/smooth`.

Result for the paper. The smooth model with the cancellation-aware discrepancy; the network ensemble and the Bayesian network as cross-checks. Draft appendix and figures: `docs/paper`.

Open items. A discrepancy correlated in m_{hh} (Gaussian process); the origin of the cancellation-dependent deviations and of the rise of ε at high mass (questions for the generator side); finer knots near threshold; use only within 250–1455 GeV.

The `dihiggs` repository at a glance



Roadmap of the documents (all in `docs/`; read top to bottom, or jump to the topic you need)

Start here	
`paper/executive_summary.pdf`	this summary
Physics and the model	
`theory.pdf`	LHC Higgs WG note on EFT descriptions of hh production (arXiv:2304.01968): the C_i
`dihiggs_model.pdf`	the `HEFTNet` model: exact θ dependence, a one-dimensional learning problem
`spectra.pdf`	one-page overview of the POWHEG spectra
Data, pipeline and checks	
`report/dihiggs_report.pdf`	the pipeline from POWHEG files to trained and tested model
`checks_plan.pdf`	three checks of the data and test pipeline
`counts/counts_note.pdf`	the spectra are scaled Poisson counts: exact noise
Training	
`huber/huber_plan.pdf`	the Huber-pull fine-tune (the accuracy gain near threshold)
`curriculum_plan.pdf, _v2.pdf`	side study: curriculum learning
The primary model	
`smooth/smooth_plan.pdf`	design of the smooth 23×80 model
`smooth/smooth_results.pdf`	its results: accuracy, calibration, within-bin and edge tests
`modelerror/model_error_note.pdf`	where the model error comes from (free-fit benchmark)
`modelerror/new_error_model.pdf`	the cancellation-aware discrepancy
Uncertainty (network studies)	
`ensemble/ensemble_plan.pdf`	seeds and bootstrap ensembles; the reference-ensemble question
`bnn/bnn_plan.pdf`	design of the Bayesian network and the HMC module
`bnn/bnn_run1.pdf–run3.pdf`	the three HMC runs (incl. a primer on mixing)
`bnn/bnn_plan2.pdf`	why HMC gave too-small widths; the Laplace plan
`bnn/laplace_explained.pdf`	the mixture of Laplace approximations, explained, with results
Paper	
`paper/appendix_draft.pdf`	draft JHEP appendix: the model and its uncertainty
`paper/fig_*.pdf`	paper figures: 9 spectra (`fig_spectra_panel`); the 23 $a_i(m_{hh})$ (`fig_coefficients`)
`paper/all_test_spectra.pdf`	all 1,000 test spectra against the smooth model (51 pages)
Practical	
`remote_gpu.pdf`	running the notebooks on the RTX 5090 (`tmux`, `nohup`)